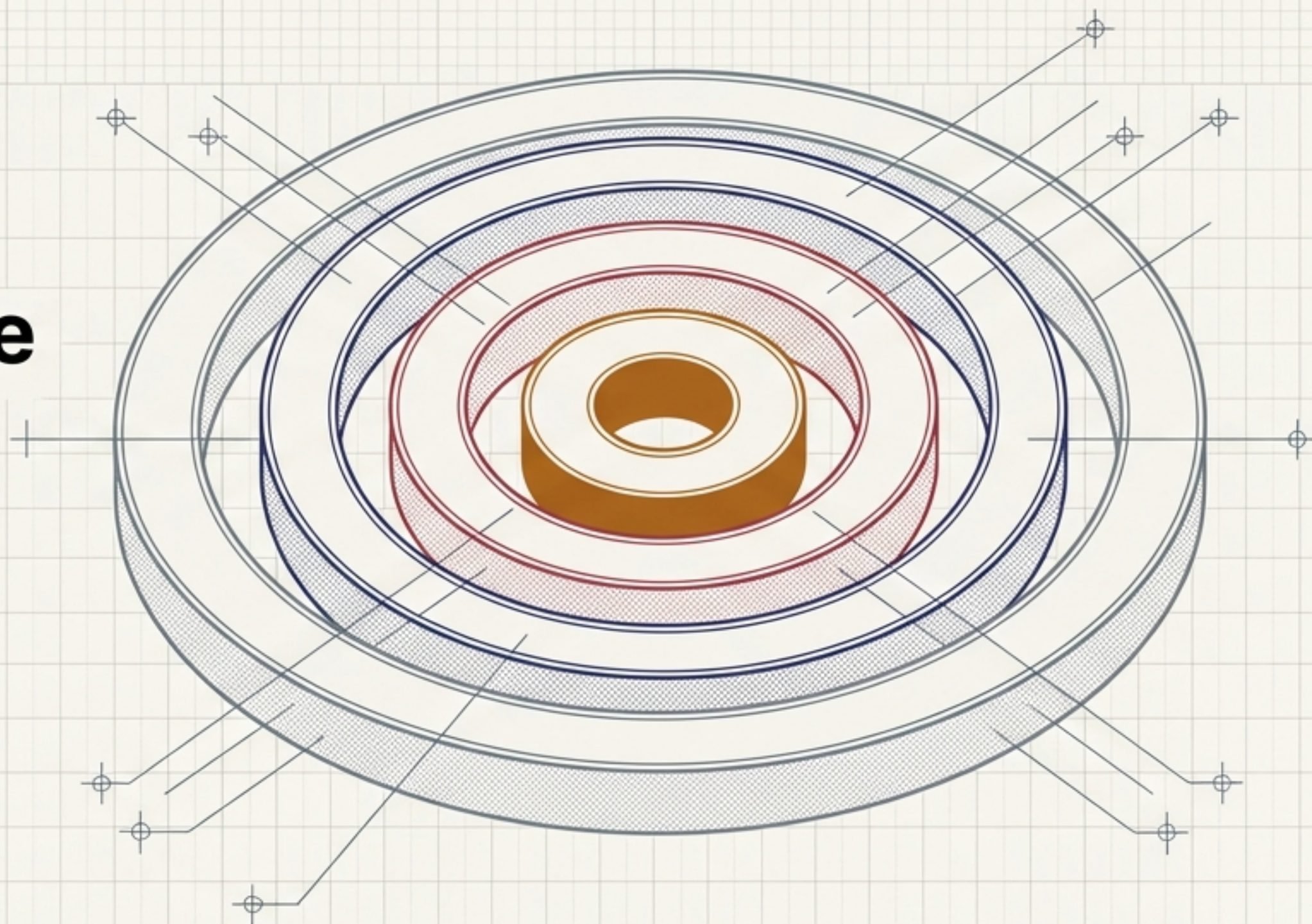


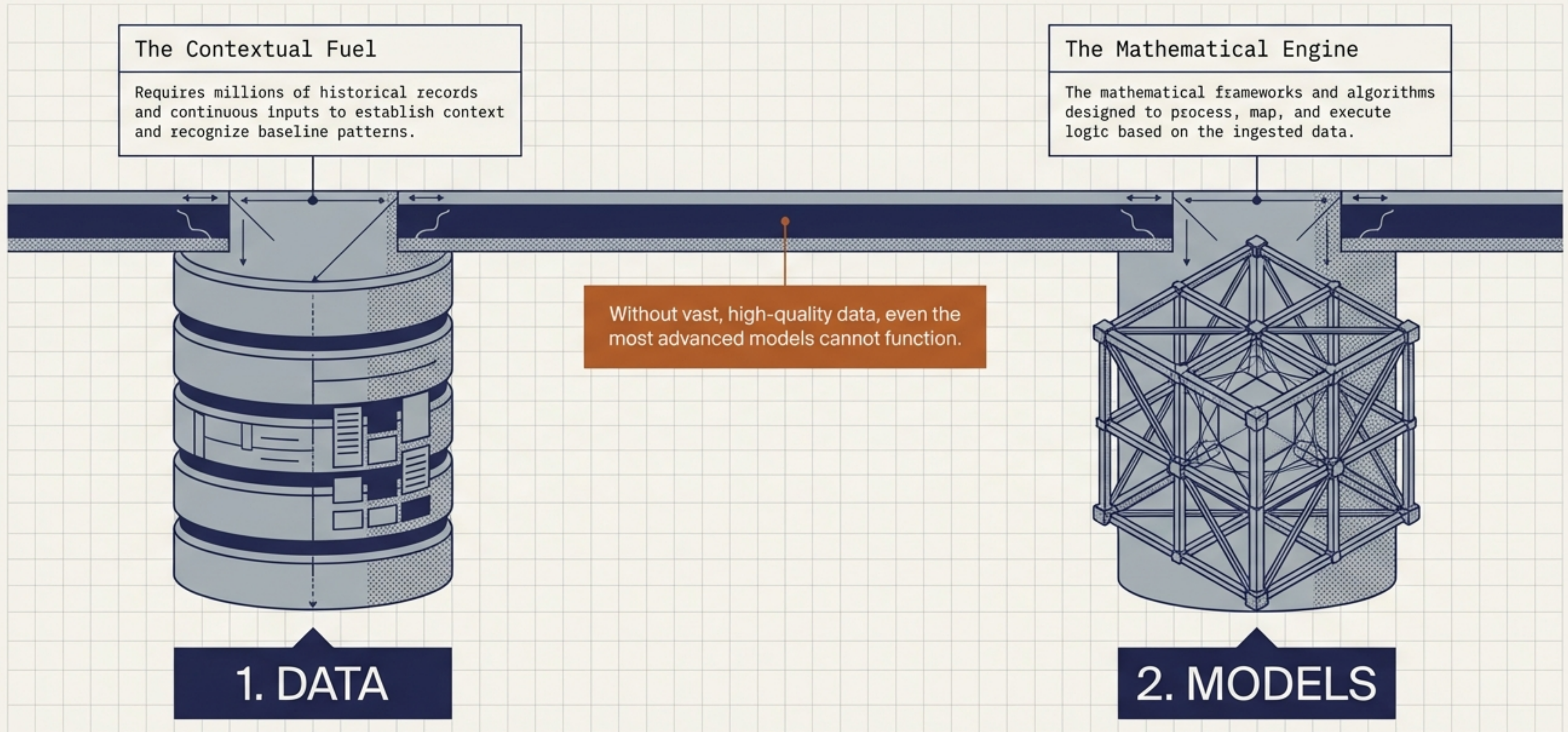
The Evolution of Artificial Intelligence

Demystifying AI, Machine Learning, Deep Learning, and Generative AI

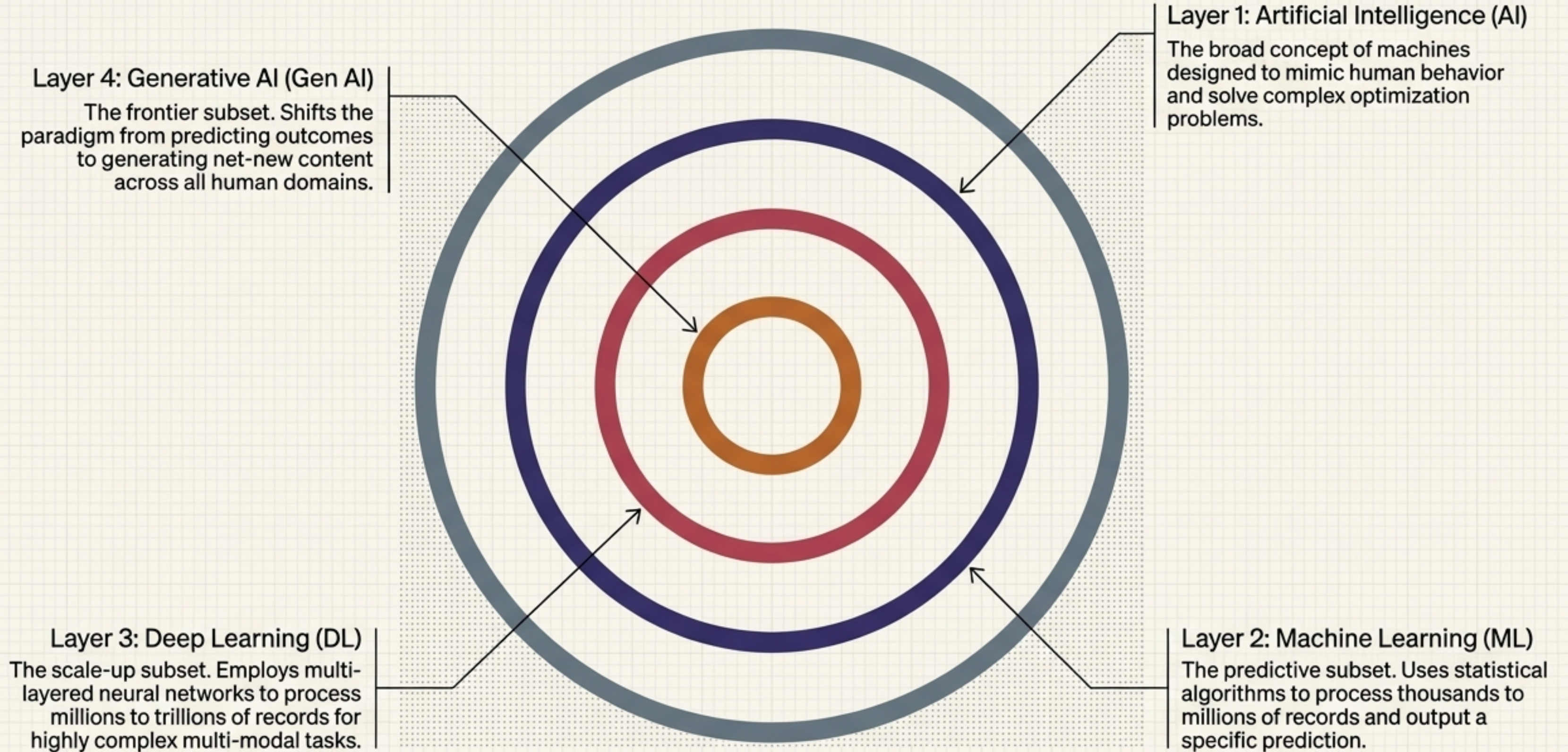
- Traditional AI
- Machine Learning
- Deep Learning
- Generative AI



The Building Blocks of Intelligence



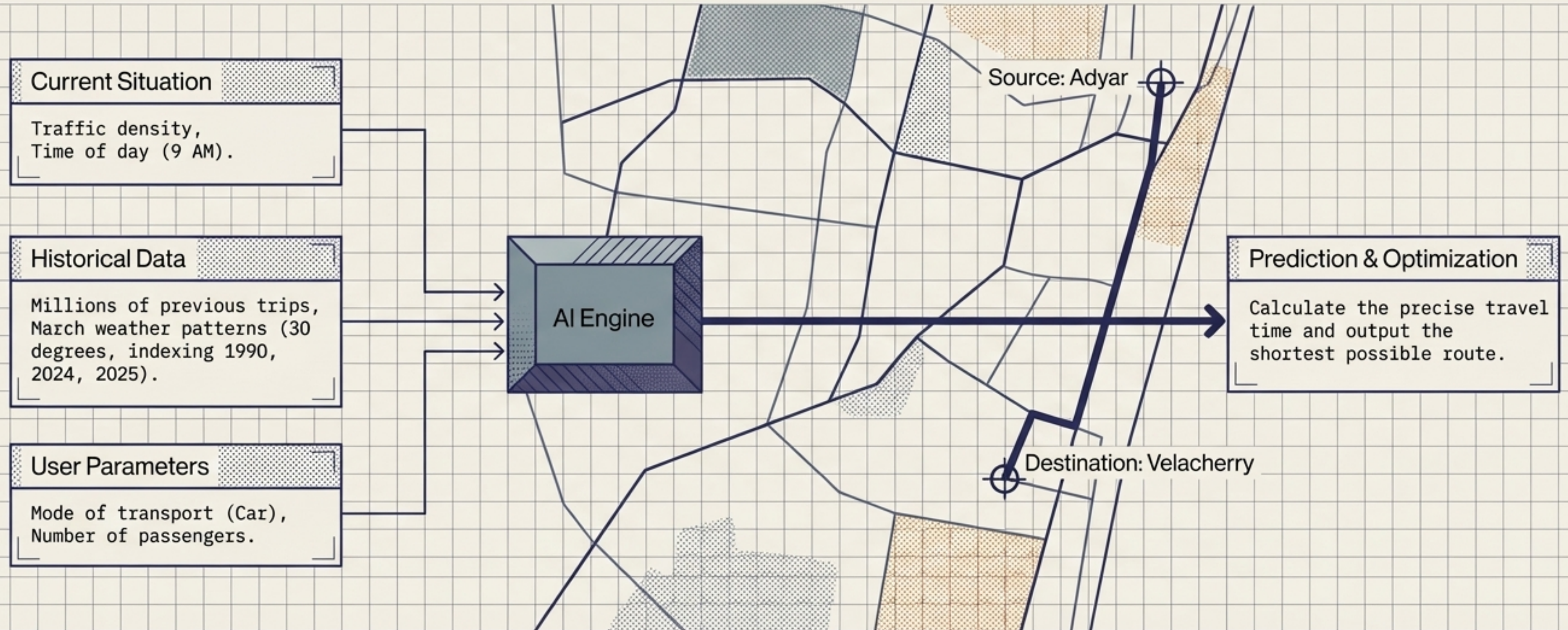
The Concentric Universe of AI



Layer 1: Traditional Artificial Intelligence

Core Function: Mimic and Optimize.

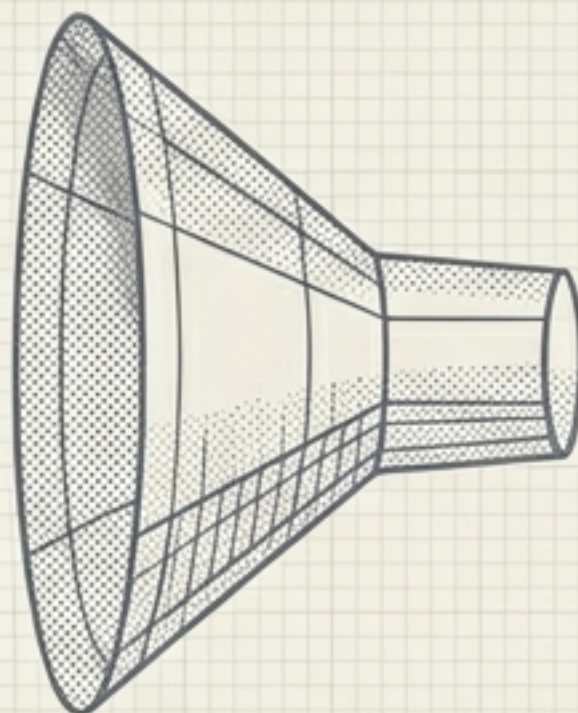
Algorithms process millions of defined variables to find the most efficient outcome.



Layer 2: Machine Learning (The Predictive Engine)

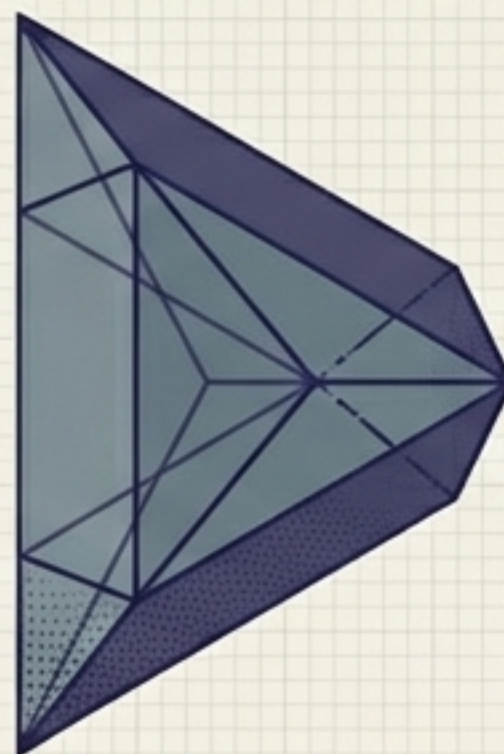
Stage 1

Input: 1,000 to Millions of Records		
Age	Blood Group	Diabetic
Blood Pressure	Location	Phone
Genetic History	Smoker	Drinker



Stage 2

Machine Learning Model



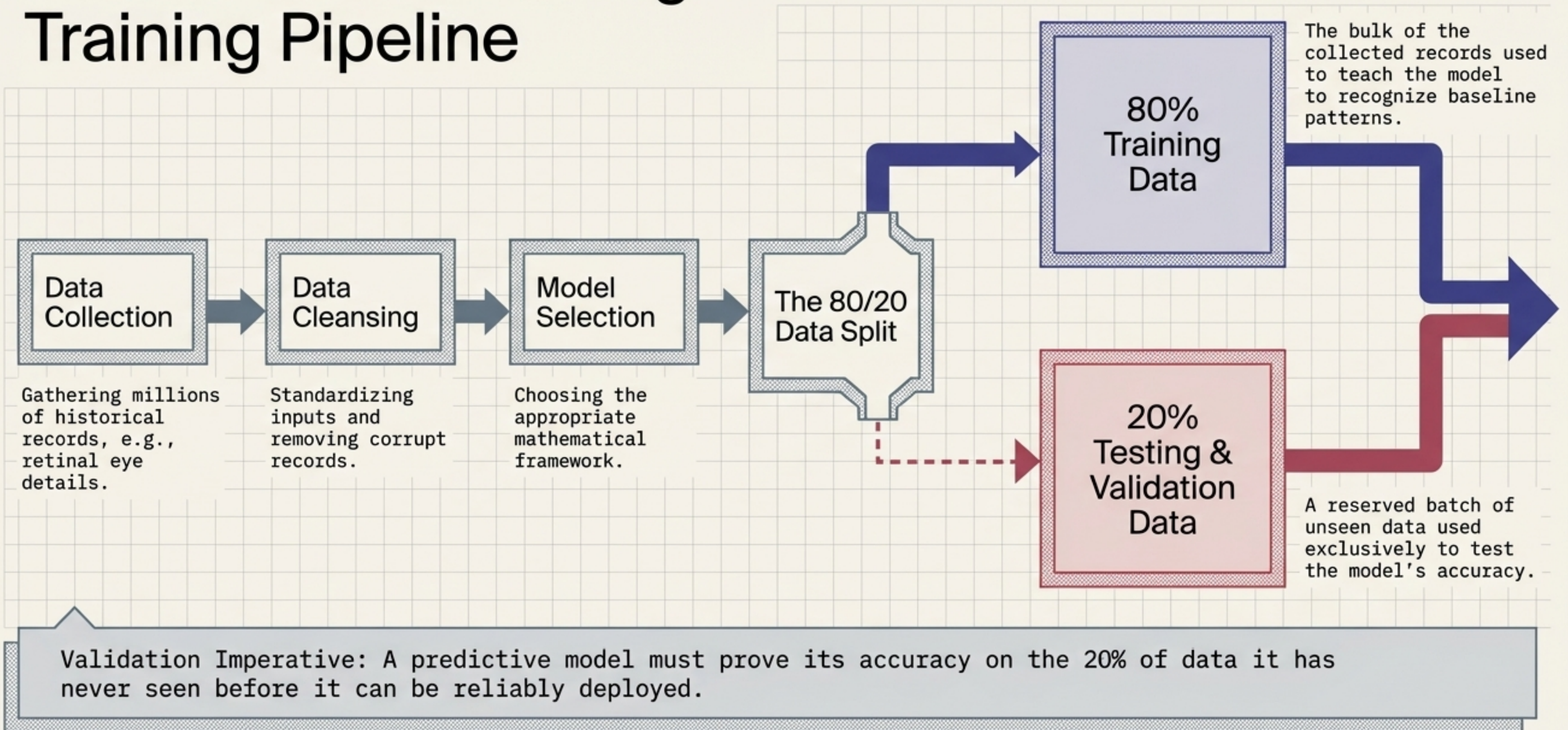
Algorithms map hidden statistical correlations within the population within the parameters without explicit programming.

Stage 3

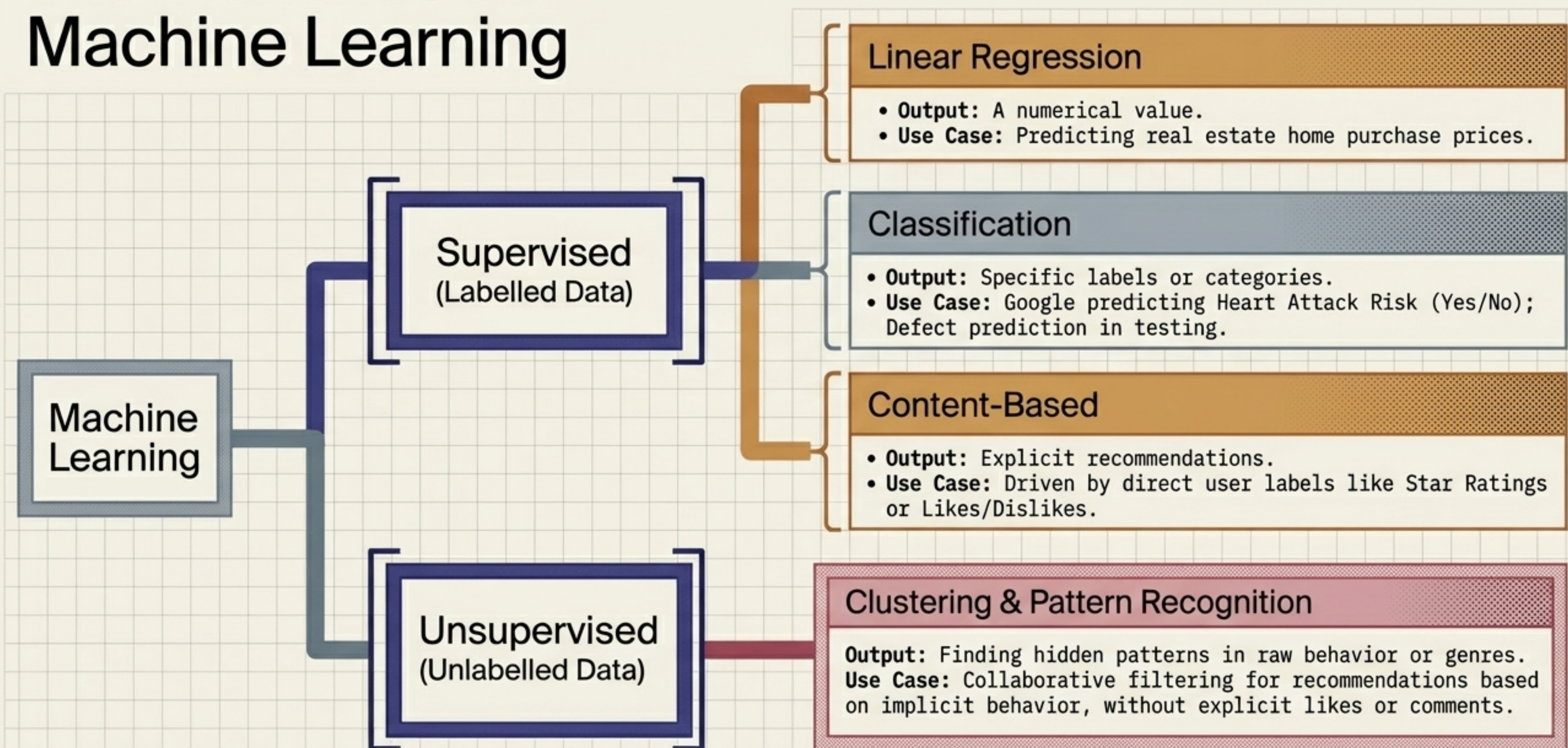
Output: Specific Prediction

Assess the statistical risk of heart disease for a new, unseen patient based on historical patterns.

The Machine Learning Training Pipeline



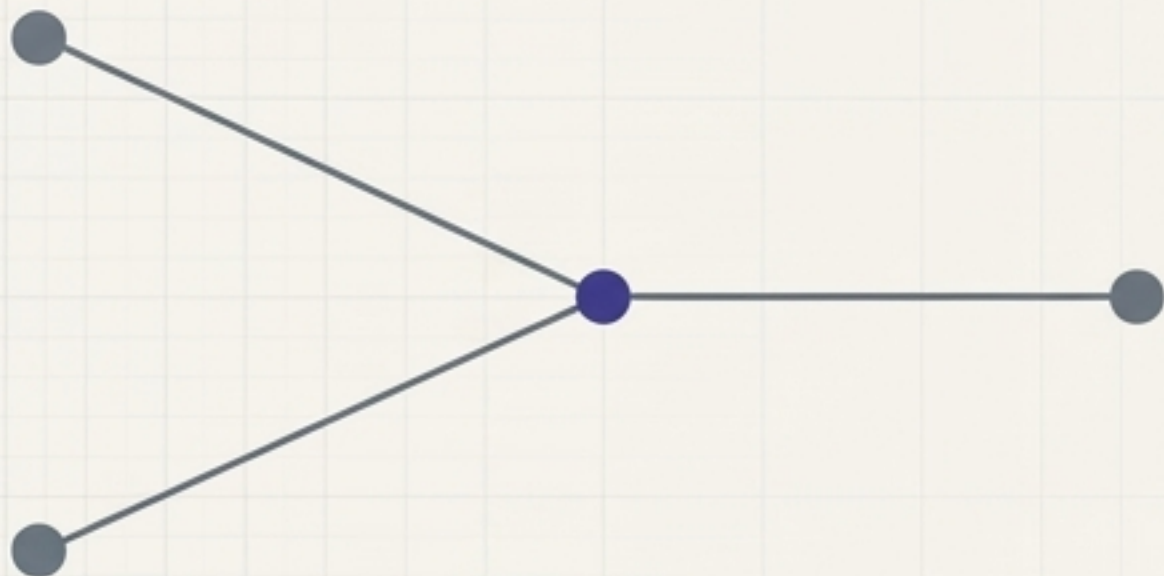
The Branches of Machine Learning



Layer 3: Deep Learning (The Scale-Up)

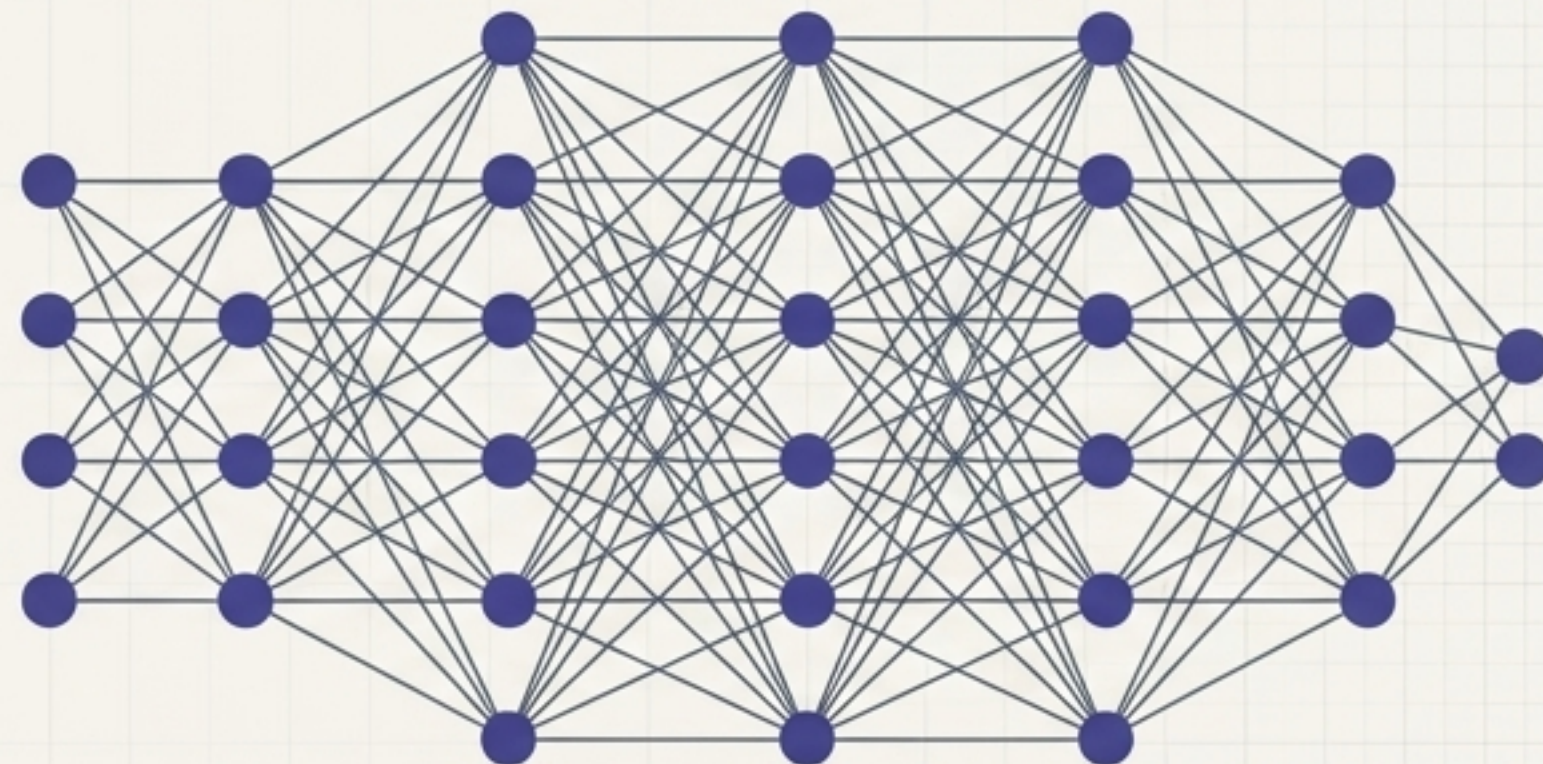
Deep Learning = Machine Learning + Neural Networks + Vast Data

Linear Processing



Handles thousands to millions of structured records.

Multi-Layered Neural Networks

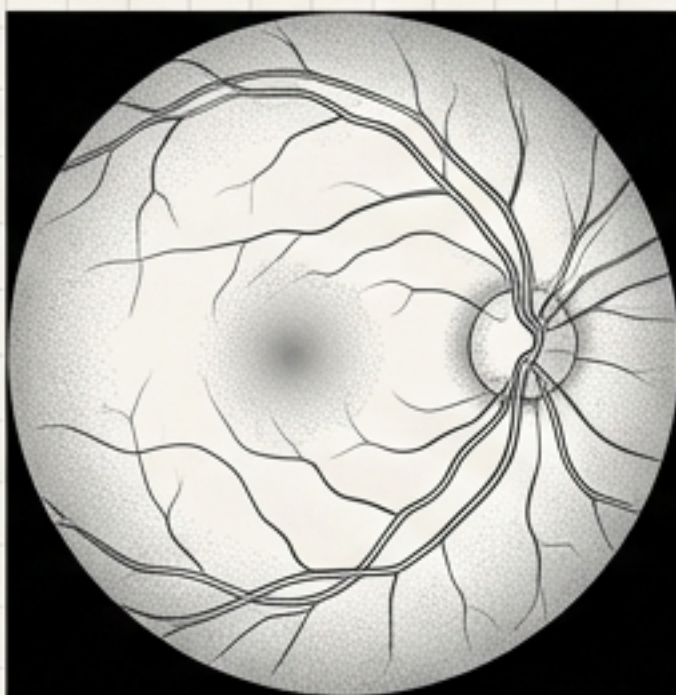


Scales to millions and trillions of records.

Algorithms are structured in continuous, hidden layers designed to process exponentially more complex, multi-modal patterns.

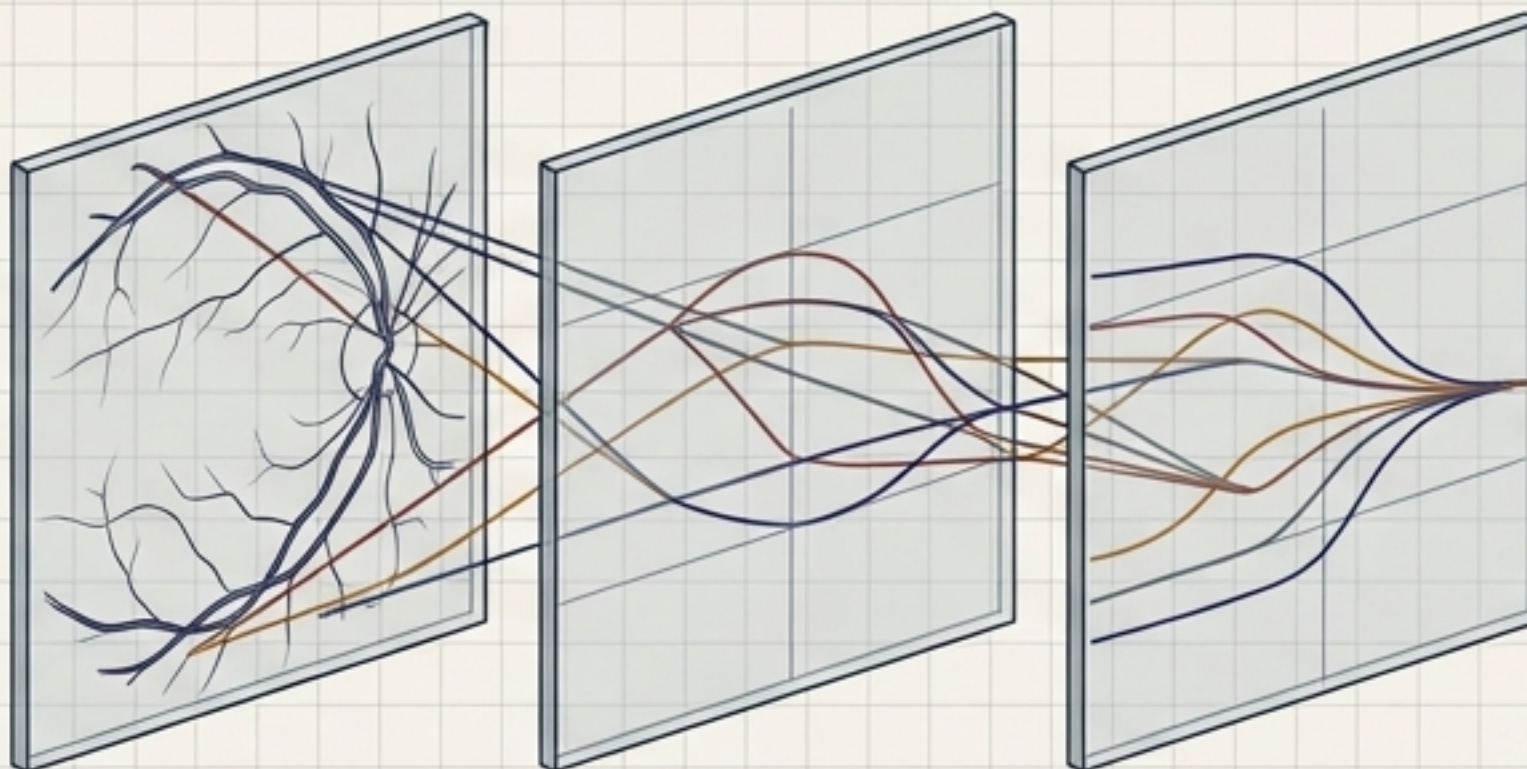
Deep Learning in Action: Computer Vision

Multi-Modal Input



A new, unseen raw retinal image.

CNN (Convolutional Neural Network)



The model, trained on millions of historical retinal images, mathematically extracts features directly from the visual data without relying on structured rows and columns.

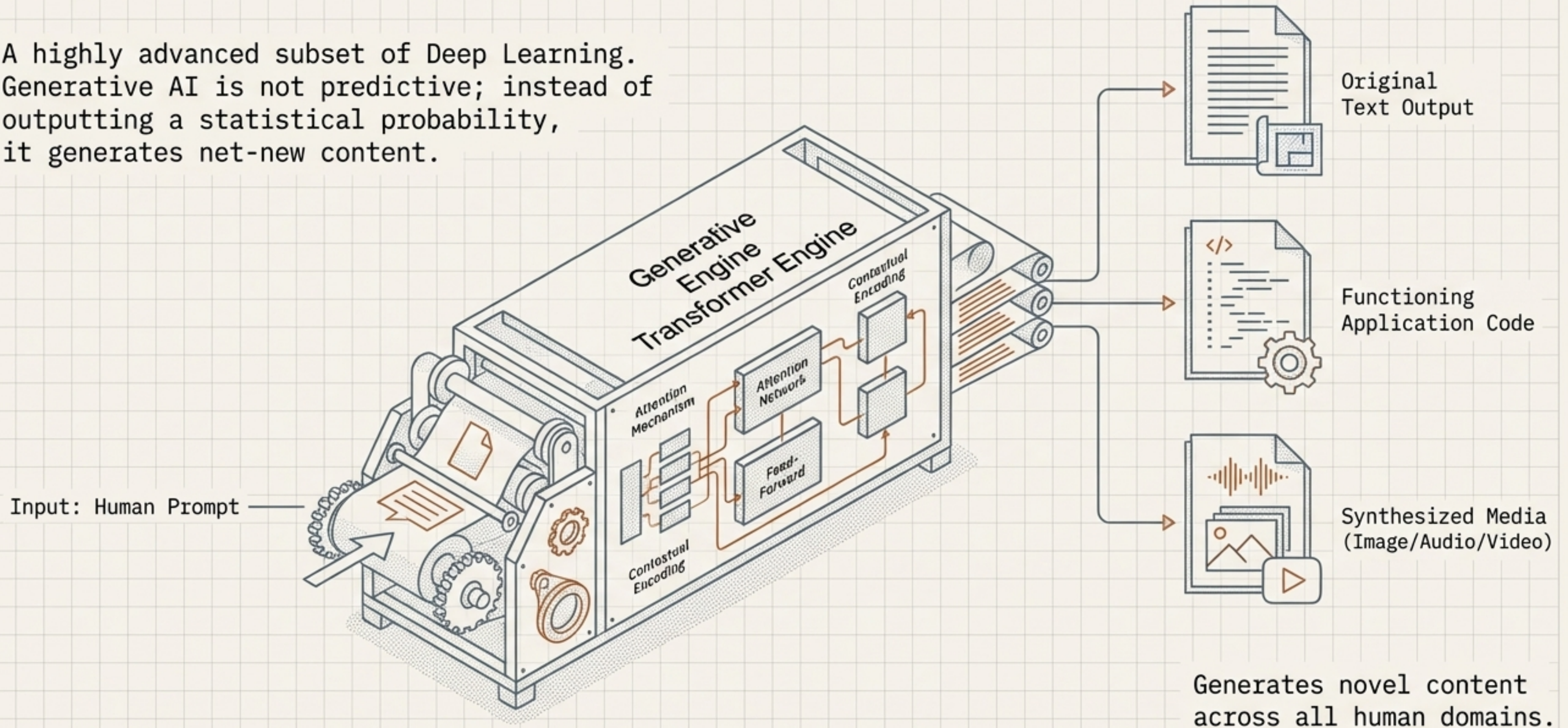
Output Probability

Predicts the highly specific statistical risk of Heart Disease.

Unlike traditional ML restricted to structured tables, Deep Learning is multi-modal, capable of natively accepting image, text, or video inputs.

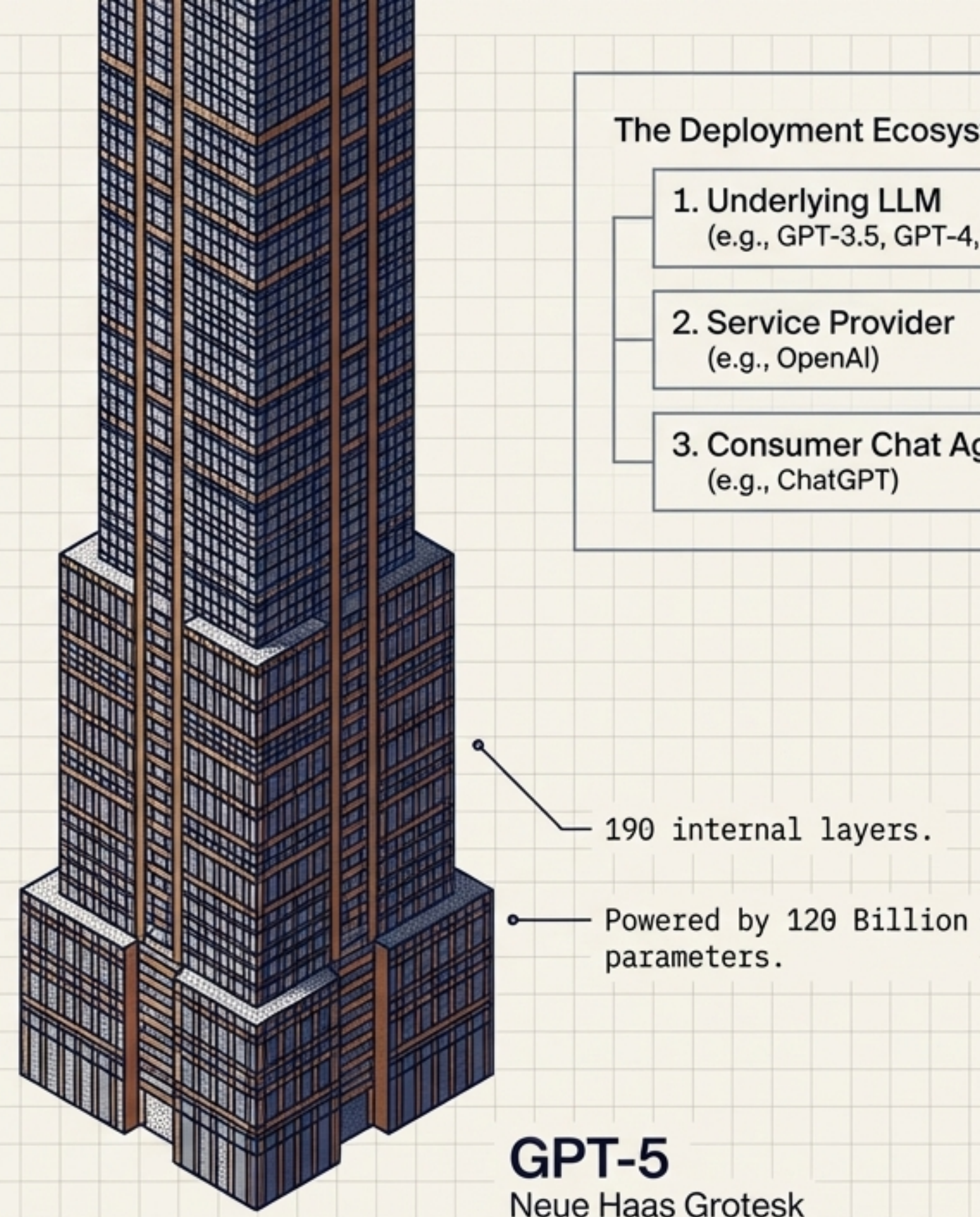
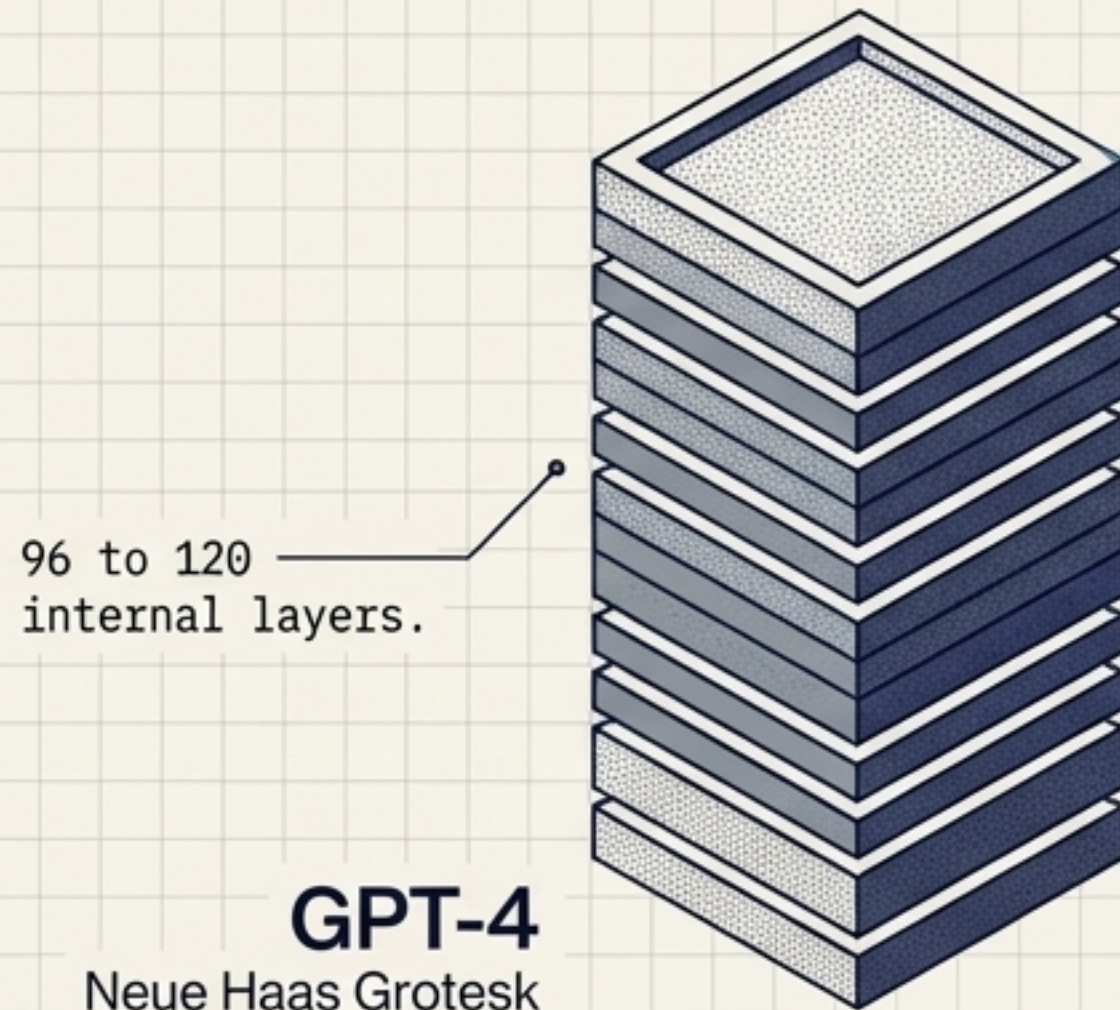
Layer 4: Generative AI (The Frontier)

A highly advanced subset of Deep Learning. Generative AI is not predictive; instead of outputting a statistical probability, it generates net-new content.



The Anatomy of Large Language Models (LLMs)

GPT (Generative Pre-trained Transformer) models are pre-trained on colossal, diverse datasets spanning all fields of human knowledge, not just software.
IBM Plex Mono

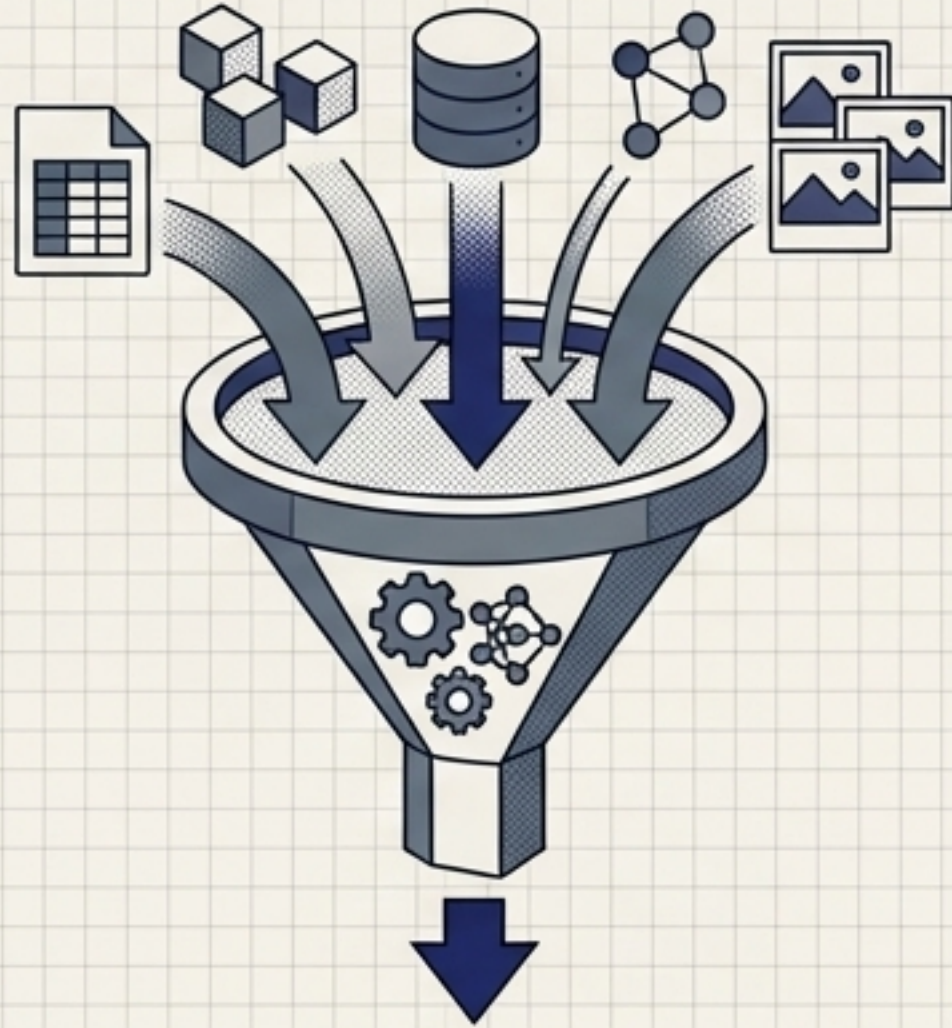


The Deployment Ecosystem:

1. Underlying LLM
(e.g., GPT-3.5, GPT-4, GPT-5)
2. Service Provider
(e.g., OpenAI)
3. Consumer Chat Agent
(e.g., ChatGPT)

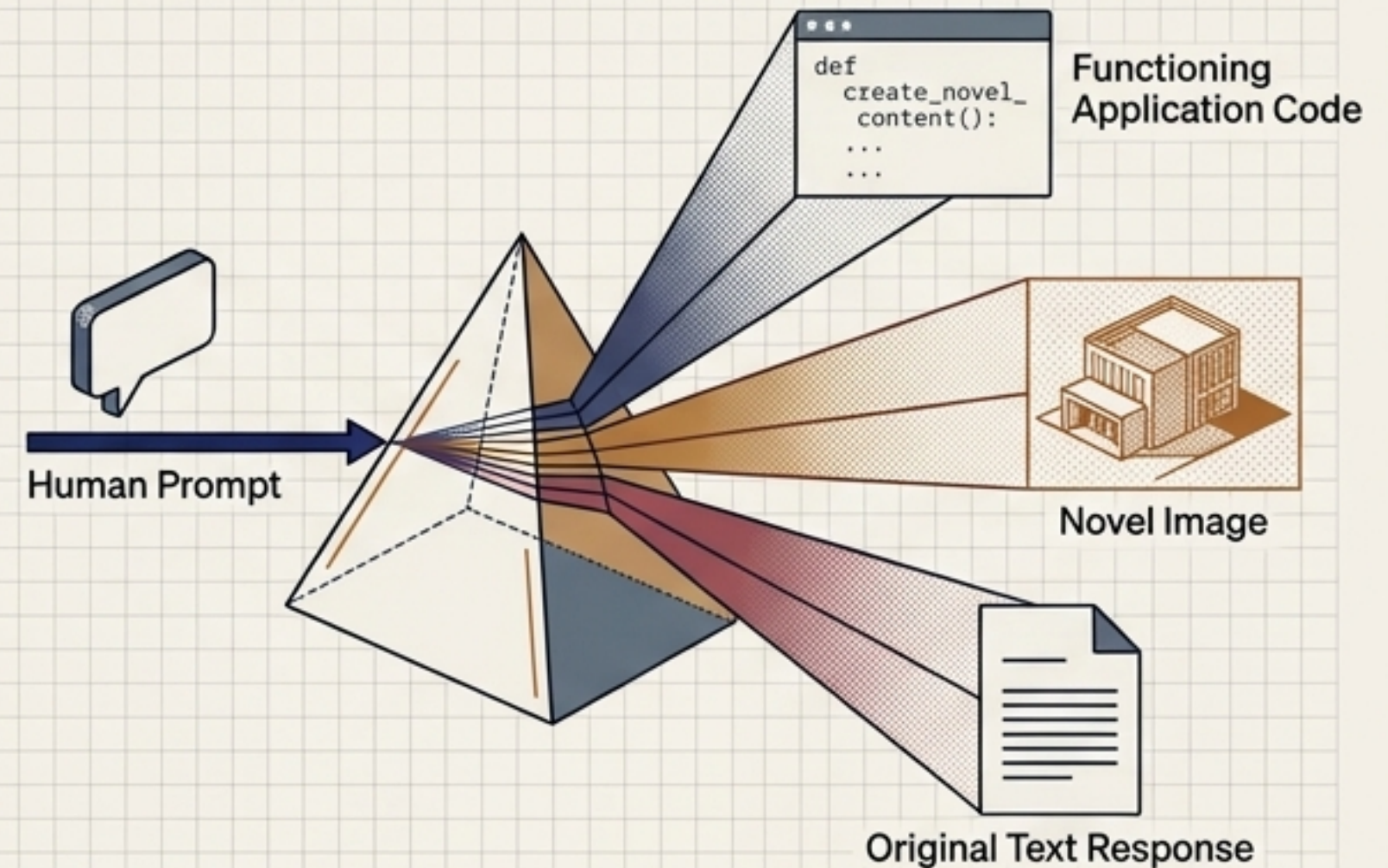
The Paradigm Shift: Predictive vs. Generative

Convergent / Predictive AI (ML & DL)



Input: Structured Data or Multi-modal Images
The Engine: Statistical Analysis & Neural Networks
Output: A Number, Category, or Label
Examples: "30 Degrees", "High Heart Disease Risk", "Yes/No"

Divergent / Generative AI



Input: Human Prompt
The Engine: Pre-trained Transformers (LLMs)
Output: Net-New Creation
Examples: A functioning block of Python code, a novel image, an original text response

The fundamental evolution from analyzing what IS to creating what COULD BE

The AI Ecosystem

Comparison Matrix

	Traditional AI	Machine Learning	Deep Learning	Generative AI
Core Action	Mimic & Optimize	Predict	Process Complexity	Generate
Data Scale	Historical patterns	Thousands to Millions	Millions to Trillions	Colossal universal datasets
Underlying Engine	Rules & Logic	Statistical Models	Neural Networks	Transformers (LLMs)
Primary Output	Optimization	Numbers / Categories	Complex classification	Net-new media & code
Use Case Example	Google Maps Routing	Heart Attack Risk (Yes/No)	CNN Retina Image Analysis	ChatGPT text generation

Key Takeaways



1. The Foundation

All modern AI is constrained by its fuel. Massive, clean historical datasets and structural mathematical models are the non-negotiable prerequisites for any intelligent system.



2. The Evolution of Prediction

Machine Learning and Deep Learning revolutionized our ability to analyze data. They extract hidden patterns to predict highly accurate outcomes—from real estate prices to complex medical diagnoses.



3. The Leap to Creation

Generative AI represents a fundamental break from prediction. Built on neural networks of unprecedented scale, it shifts the paradigm to generating original content and code across all human domains.

Looking Ahead

Next Module: Day 2 — LLM Anatomy

A deep dive into the internal working, architecture, and parameter tuning of Large Language Models.

